

HEEJUN KIM

 [Homepage](#)  [Email](#)  [GitHub](#)  [Google Scholar](#)

EDUCATION

- Korea Advanced Institute of Science and Technology (KAIST)** *September 2026 - present*
M.S. in Artificial Intelligence
- Gwangju Institute of Science and Technology (GIST)** *February 2020 - August 2026*
B.S. in Electrical Engineering and Computer Science, GPA: 4.2/4.5 (magna cum laude)
- University of California, Berkeley (UC Berkeley)** *June 2024 - August 2024*
Summer Session, GPA 4.0/4.0

RESEARCH EXPERIMENTS

- Scalable and Efficient AI Lab** *September 2026 - present*
Korea Advanced Institute of Science and Technology, Advisor: [Prof. Sehoon Kim](#) *Seoul, South Korea*
· Improving reasoning capabilities and accelerating inference in recursive architectures
- Learning and Adaptation Algorithm Lab** *January 2026 - August 2026*
Seoul National University, Advisor: [Prof. Taesup Kim](#) *Seoul, South Korea*
· Conducting research on Agentic Memory Using LLM Uncertainty Estimation
· Studied internal behavioral mechanisms of LLMs underlying reasoning, hallucination, and memory recall
- Data Science Lab** *Feb 2025 - August 2026, January 2024 - June 2024*
Gwangju Institute of Science and Technology, Advisor: [Prof. Sundong Kim](#) *Gwangju, South Korea*
· Researched test-time adaptation with reinforcement learning for self-improving reasoning capabilities of large language models on the Abstract Reasoning Corpus (ARC)
· Investigated metacognition-driven strategies for helping large language models escape local minima in reasoning
- Forschungszentrum Jülich** *September 2024 - February 2025*
Korea-Germany Junior Research Fellowship Program 2024 *Jülich, Germany*
· Host Institute: INM-7 directed by [Prof. Simon B. Eickhoff](#) and contributed to Mathematical Neuroscience Group led by [Dr. Oleksandr Popovych](#).
· Developed preprocessing pipeline for multi-modal MRI human connectome data using HPC and HyperThreading (Project Supervisor: [Dr. Kyesam Jung](#))

PUBLICATIONS

- From Noise to Diversity: Random Embedding Injection in LLM Reasoning**
Heejun Kim, Seungpil Lee, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, Sundong Kim
NeurIPS 2026, ICML 2026 Mechanistic Interpretability Workshop (Spotlight, 2.8% acceptance rate). [\[Paper\]](#)
- Hallucination as Commitment Failure: Larger LLMs Misfire Despite Knowing the Answer**
Jewon Yeom, Jaewon Sok, [Heejun Kim](#), Seonghyeon Park, Jeongjae Park, Taesup Kim
NeurIPS 2026, ICML 2026 Mechanistic Interpretability Workshop. [\[Paper\]](#)
- On the Role of Proposal Support in Diffusion-Based Offline RL for Sequential Decision-Making**
Yunho Kim, Jaehyun Park, [Heejun Kim](#), Sejin Kim, Byung-Jun Lee, Sundong Kim
ICML 2026 DEMO Workshop. [\[Paper\]](#)

PROJECTS

- Skill-based Reinforcement Learning with UCB Skill Selection** *May 2025 - June 2025*
· Reproduced BeCL framework for unsupervised skill discovery in reinforcement learning environments
· Developed Multi-Armed Bandit approach with UCB algorithm to improve downstream task adaptation efficiency. [\[Poster\]](#)

Hyundai AI Challenge

September 2023 - October 2023

- Processed large-scale port logistics datasets (cleaning, feature extraction, temporal aggregation).
- Built tree-based boosting models to predict ship waiting times.
- Ranked Top 25/330 teams among 1,300+ participants.

LEADERSHIPS

Teaching Assistant

Fall 2025

- Machine Learning & Deep Learning (AI5213/EC4213): led TA sessions, held office hours, managed assignments/exams, and organized Kaggle competition. [\[Homepage\]](#)

Korea Army Instructor

February 2021 - August 2022

- Served as an Army instructor in the 50th Division in Daegu, South Korea
- Conducted basic training for approximately 2000 military personnel

EXTRA ACTIVITIES

Google Developer Student Club

September 2023 - February 2024

- Participated in an AI study focused on machine learning and deep learning
- Studied models in ML/DL and analysis of top team models in Kaggle AI competitions
- Established study directions and organized study materials as a team leader

Financial Investment Conference

March 2023 - March 2024

- Conducted a study on economics
- Analyzed and presented economic markets on a weekly basis

GIST Club Federation

February 2020 - December 2020

- Managed clubs in Gwangju Institute of Science and Technology

SKILLS

Programming Languages	Python, C/C++
Python Packages & Framework	PyTorch, VeRL, Scikit-learn, Pandas, Numpy, Matplotlib, Ray
Operating System	Linux, Windows, MacOS
Others	Docker, Wandb, SLURM, HTCondor, SQLite

HONORS

Government Scholarship, Gwangju Institute of Science and Technology	2020 - August 2026
Government Scholarship, STEM scholarship	2020 - August 2026
Dean's List, College of Information and Computing, GIST	Spring 2025, Fall 2025
GIST International Internship Support Program	September 2024 - February 2025
Korea-Germany Junior Research Fellowship Program 2024	September 2024 - February 2025
· Organized by the Max Planck POSTECH/KOREA Research (MPK)	
· Fund: 12,000,000 KRW (8,197 EUR) by National Research Foundation of Korea (NRF), No. RS-2023-00254343.	